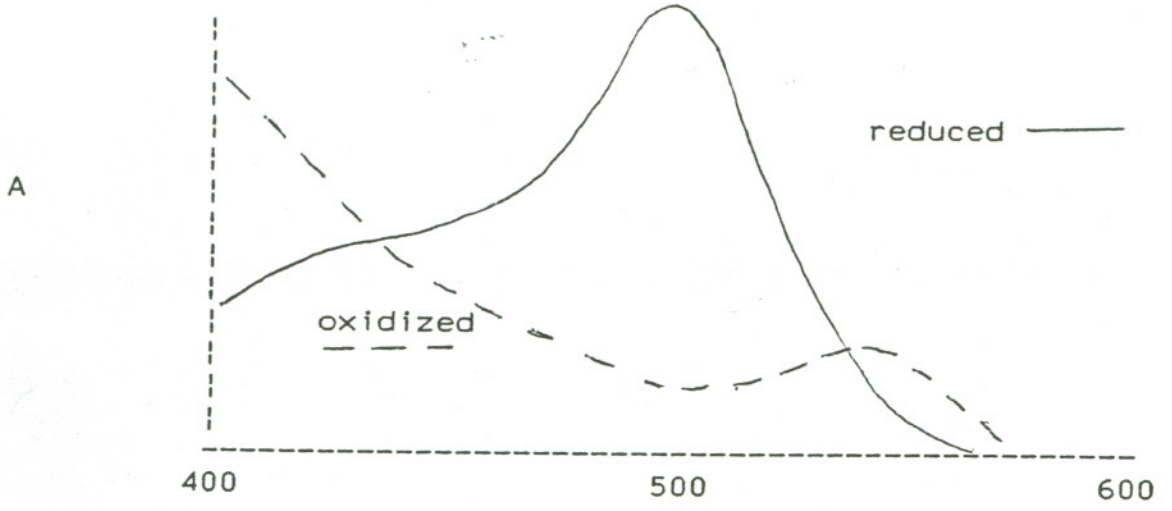
BIOCHEMISTRY CUMULATIVE EXAM

December 7, 1991

You have isolated an new electron transfer protein from a bacterium which has been found growing on the seats in the football stadium. This protein has the following spectra in its fully oxidized and fully reduced states.



A 1 mM solution of the protein has an absorbance of 0.25 at 500 nm when fully oxidized and an absorbance of 1.75 when fully reduced.

You are now attempting to further characterize the redox properties of this protein. You have prepared 1 mM solutions of the protein and equilibrated them against an electrode with the following potentials and measured the absorbance at 500 nm. The following data has been collected.

Eh (V vs SHE*)	A at 500 nm
0.391	1.25
0.409	0.75
0.418	0.55
0.427	0.42
0.432	0.37
0.436	0.34
0.438	0.325

*SHE = standard hydrogen electrode

Use these data to graphically determine E° and n for this newly discovered protein. A piece of graph paper is provided. Show calculations which allow you to get from the raw data to the information plotted on the graph.